

IMAGE PROCESSING USING MATLAB:

Spatial Filtering



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Spatial filtering techniques refer to those operations where the resulting value of a pixel at a given coordinate is a function of the original pixel value at that point as well as the original pixel value of some of its neighbours. These operations are classified into linear filtering operations and non-linear filtering operations.

In linear filters, the resulting output pixel is computed as a sum of products of the pixel values and mask coefficients in the pixel's neighbourhood in the original image. In other words, if the function by which the new grey value is calculated is a linear function of all the grey values in the mask, then the filter is referred to as a linear filter. Mean filter is an example of linear filter.

In non-linear filters, the resulting output pixel is selected from an ordered sequence of pixel values in the pixel's neighbourhood in the original image. Median filter is an example of non-linear filter.

Linear filters

Convolution and correlation are the two fundamental mathematical operations involved in linear filters based on neighbourhood-oriented image processing algorithms.

Convolution. Convolution processes an image by computing, for each pixel, a weighted sum of the values of that pixel and its neighbours. Depending on the choice of weights, a wide variety of image processing operations can be implemented.

Different convolution masks produce different results when applied to the same input image. These operations are referred to as filtering operations and the masks as spatial filters. Spatial filters are often named based on their behaviour in the spatial frequency. Low-pass filters (LPFs) are those spatial filters whose effect on the output image is equivalent to attenu-

ating the high-frequency components (fine details in the image) and preserving the low-frequency components (coarser details and homogeneous areas in the image). These filters are typically used to either blur an image or reduce the amount of noise present in the image. Linear low-pass filters can be implemented using 2D convolution masks with non-negative coefficients.

High-pass filters (HPFs) work in a complementary way to LPFs, that is, these preserve or enhance high-frequency components with the possible side-effect of enhancing noisy pixels as well. High-frequency components include fine details, points, lines and edges. In other words, these highlight transitions in intensity within the image.

There are two in-built functions in MATLAB's Image Processing Toolbox (IPT) that can be used to implement 2D convolution: `conv2` and `filter2`.

`conv2` computes 2D convolution between two matrices. For example, `C = conv2(A,B)` computes the two-dimensional convolution of matrices A and B. If one of these matrices describes a two-dimensional finite impulse response (FIR) filter, the other matrix is filtered in two dimensions.

`filter2` function rotates the convolution mask, that is, 2D FIR filter, by 180° in each direction to create a convolution kernel and then calls `conv2` to perform the convolution operation.

`Y = filter2(h,X)` filters the data in 'X' with the two-dimensional FIR filter in the matrix 'h'. It computes the result 'Y' using two-dimensional correlation, and returns the central part of the correlation that is of the same size as 'X'.

`Y = filter2(h,X,shape)` returns the part of 'Y' specified by the shape parameter. Shape is a string with one of these values:

1. 'full' returns the full two-dimen-